

Network Sniffing

Adversaries may passively sniff network traffic to capture information about an environment, including authentication material passed over the network. Network sniffing refers to using the network interface on a system to monitor or capture information sent over a wired or wireless connection. An adversary may place a network interface into promiscuous mode to passively access data in transit over the network, or use span ports to capture a larger amount of data.

Data captured via this technique may include user credentials, especially those sent over an insecure, unencrypted protocol. Techniques for name service resolution poisoning, such as [Name Resolution Poisoning and SMB Relay](#), can also be used to capture credentials to websites, proxies, and internal systems by redirecting traffic to an adversary.

Network sniffing may reveal configuration details, such as running services, version numbers, and other network characteristics (e.g. IP addresses, hostnames, VLAN IDs) necessary for subsequent [Lateral Movement](#) and/or [Stealth](#) activities. Adversaries may likely also utilize network sniffing during [Adversary-in-the-Middle](#) (AiTM) to passively gain additional knowledge about the environment.

In cloud-based environments, adversaries may still be able to use traffic mirroring services to sniff network traffic from virtual machines. For example, AWS Traffic Mirroring, GCP Packet Mirroring, and Azure vTap allow users to define specified instances to collect traffic from and specified targets to send collected traffic to. [\[1\]\[2\]\[3\]](#) Often, much of this traffic will be in cleartext due to the use of TLS termination at the load balancer level to reduce the strain of encrypting and decrypting traffic. [\[4\]\[5\]](#) The adversary can then use exfiltration techniques such as Transfer Data to Cloud Account in order to access the sniffed traffic. [\[4\]](#)

On network devices, adversaries may perform network captures using [Network Device CLI](#) commands such as `monitor capture`. [\[6\]\[7\]](#)

ID: T1040

Sub-techniques: No sub-techniques

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Tactics: [Credential Access](#), [Discovery](#)

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Platforms: IaaS, Linux, Network Devices, Windows, macOS

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Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team used BlackEnergy 's network sniffer module to discover user credentials being sent over the network between the local LAN and the power grid's industrial control systems. [8]
G0007	APT28	APT28 deployed the open source tool Responder to conduct NetBIOS Name Service poisoning, which captured usernames and hashed passwords that allowed access to legitimate credentials. [9] [10] APT28 close-access teams have used Wi-Fi pineapples to intercept Wi-Fi signals and user credentials. [11]
G0064	APT33	APT33 has used SniffPass to collect credentials by sniffing network traffic. [12]
C0046	ArcaneDoor	ArcaneDoor included network packet capture and sniffing for data collection in victim environments. [13] [14]
S1224	CASTLETAP	CASTLETAP has the ability to create a raw promiscuous socket to sniff network traffic. [15]
S1204	cd00r	cd00r can use the libpcap library to monitor captured packets for specific sequences. [16]
G0105	DarkVishnya	DarkVishnya used network sniffing to obtain login data. [17]
S0367	Emotet	Emotet has been observed to hook network APIs to monitor network traffic. [18]
S0363	Empire	Empire can be used to conduct packet captures on target hosts. [19]
S0661	FoggyWeb	FoggyWeb can configure custom listeners to passively monitor all incoming HTTP GET and POST requests sent to the AD FS server from the intranet/internet and intercept HTTP requests that match the custom URI patterns defined by the actor. [20]
S0357	Impacket	Impacket can be used to sniff network traffic via an interface or raw socket. [21]
S1203	J-magic	J-magic has a pcap listener function that can create an Extended Berkley Packet Filter (eBPF) on designated interfaces and ports. [22]
S1206	JumbledPath	JumbledPath has the ability to perform packet capture on remote devices via actor-defined jump-hosts. [23]

ID	Name	Description
G0094 Kimsuky		Kimsuky has used the Nirsoft SniffPass network sniffer to obtain passwords sent over non-secure protocols. [24] [25]
S1186 Line Dancer		Line Dancer can create and exfiltrate packet captures from compromised environments. [13]
S0443 MESSAGETAP		MESSAGETAP uses the libpcap library to listen to all traffic and parses network protocols starting with Ethernet and IP layers. It continues parsing protocol layers including SCTP, SCCP, and TCAP and finally extracts SMS message data and routing metadata. [26]
S0590 NBTscan		NBTscan can dump and print whole packet content. [27] [28]
S0587 Penguin		Penguin can sniff network traffic to look for packets matching specific conditions. [29] [30]
S0378 PoshC2		PoshC2 contains a module for taking packet captures on compromised hosts. [31]
C0056 RedPenguin		During RedPenguin , UNC3886 used a passive backdoor to act as a libpcap-based packet sniffer. [32]
S0019 Regin		Regin appears to have functionality to sniff for credentials passed over HTTP, SMTP, and SMB. [33]
S0174 Responder		Responder captures hashes and credentials that are sent to the system after the name services have been poisoned. [34]
G1045 Salt Typhoon		Salt Typhoon has used a variety of tools and techniques to capture packet data between network interfaces. [23]
G0034 Sandworm Team		Sandworm Team has used interceptor-NG to sniff passwords in network traffic. [35]
S9024 SPAWNCHIMERA		SPAWNCHIMERA has monitored and filtered network traffic on compromised edge devices, allowing legitimate traffic to pass while redirecting attacker-controlled traffic to infrastructure under adversary control. [36] [37]
G1048 UNC3886		UNC3886 has used the LOOKOVER sniffer to sniff TACACS+ authentication packets. [38]
G1047 Velvet Ant		Velvet Ant has used a custom tool, "VELVETTAP", to perform packet capture from compromised F5 BIG-IP devices. [39]

ID	Name	Description
S1154	VersaMem	VersaMem hooked the Catalina application filter chain <code>doFilter</code> on compromised systems to monitor all inbound requests to the local Tomcat web server, inspecting them for parameters like passwords and follow-on Java modules. ^[40]

Mitigations

ID	Mitigation	Description
M1041	Encrypt Sensitive Information	Ensure that all wired and/or wireless traffic is encrypted appropriately. Use best practices for authentication protocols, such as Kerberos, and ensure web traffic that may contain credentials is protected by SSL/TLS.
M1032	Multi-factor Authentication	Use multi-factor authentication wherever possible.
M1030	Network Segmentation	Deny direct access of broadcasts and multicast sniffing, and prevent attacks such as Name Resolution Poisoning and SMB Relay
M1018	User Account Management	In cloud environments, ensure that users are not granted permissions to create or modify traffic mirrors unless this is explicitly required.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0314	Detection Strategy for Network Sniffing Across Platforms	AN0875	Detects suspicious execution of network monitoring tools (e.g., Wireshark, tshark, Microsoft Message Analyzer), driver loading indicative of promiscuous mode, or non-admin user privilege escalation to access NICs for capture.
		AN0876	Correlates interface mode changes to promiscuous with execution of sniffing tools like tcpdump, tshark, or custom pcap libraries. Detects abnormal NIC configurations and unauthorized sniffing from non-root sessions.
		AN0877	Detects enabling of interface sniffing via packet capture tools or AppleScript triggering tcpdump. Leverages Unified Logs and process lineage to identify suspicious use of <code>pfctl</code> , <code>tcpdump</code> , or <code>libpcap</code> libraries.
		AN0878	Detects creation of traffic mirroring sessions (e.g., AWS VPC Traffic Mirroring, Azure vTAP) that redirect traffic from critical assets to other virtual instances, often followed by file creation or session establishment.

ID	Name	Analytic ID	Analytic Description
		AN0879	Detects execution of capture commands via CLI (monitor capture, debug packet, etc.) or unauthorized CLI access followed by logging configuration changes on Cisco/Juniper/Arista gear.

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